Paternity Table Sources

Bibliography of studies included in Rosenbaum, Grebe, & Silk (2026)

March 05, 2026

References

Alberts, S. C., Buchan, J. C., & Altmann, J. (2006). Sexual selection in wild baboons: From mating opportunities to paternity success. *Animal Behaviour*, *72*, 1177–1196. https://doi.org/10.1016/j.anbehav.2006.05.001

Altmann, J., Alberts, S. C., Haines, S. A., Dubach, J., Muruth, P., Coote, T., Geffen, E., Cheesman, D. J., Mututua, R. S., Saiyalel, S. N., Wayne, R. K., Lacy, R. C., & Bruford, M. W. (1996). Behavior predicts genetic structure in a wild primate group. *Proceedings of the National Academy of Sciences of the United States of America*, *93*, 5797–5801. https://doi.org/10.1073/pnas.93.12.5797

Banes, G. L., Galdikas, B. M. F., & Vigilant, L. (2015). Male orang-utan bimaturism and reproductive success at Camp Leakey in Tanjung Puting National Park, Indonesia. *Behavioral Ecology and Sociobiology*, *69*, 1785–1794. https://doi.org/10.1007/s00265-015-1991-0

Barelli, C., Matsudaira, K., Wolf, T., Roos, C., Heistermann, M., Hodges, K., Ishida, T., Malaivijitnond, S., & Reichard, U. H. (2013). Extra-pair paternity confirmed in wild white-handed gibbons. *American Journal of Primatology*, *75*, 1185–1195. https://doi.org/10.1002/ajp.22180

Boesch, C., Kohou, G., Néné, H., & Vigilant, L. (2006). Male competition and paternity in wild chimpanzees of the Taï Forest. *American Journal of Physical Anthropology*, *130*, 103–115. https://doi.org/10.1002/ajpa.20341

Bonadonna, G., Torti, V., DeGregorio, C., Valente, D., Randrianarison, R. M., Pozzi, L., Gamba, M., & Giacoma, C. (2019). Evidence of genetic monogamy in the lemur Indri (Indri indri). *American Journal of Primatology*. https://doi.org/10.1002/ajp.22993

Borries, C. (n.d.). *Unpublished paternity data for Phayre's leaf monkeys.*

Bradley, B. J., Doran-Sheehy, D. M., Lukas, D., Boesch, C., & Vigilant, L. (2004). Dispersed male networks in western gorillas. *Current Biology*, *14*, 510–513. https://doi.org/10.1016/j.cub.2004.02.062

Bradley, B. J., Robbins, M. M., Williamson, E. A., Steklis, H. D., Steklis, N. G., Eckhardt, N., Boesch, C., & Vigilant, L. (2005). Mountain gorilla tug-of-war: Silverbacks have limited control over reproduction in multimale groups. *Proceedings of the National Academy of Sciences of the United States of America*, *101*, 9187–9191. https://doi.org/10.1073/pnas.0502019102

Brauch, K., Hodges, K., Engelhardt, A., Fuhrmann, K., Shaw, E., & Heistermann, M. (2008). Sex-specific reproductive behaviours and paternity in free-ranging Barbary macaques (Macaca sylvanus). *Behavioral Ecology and Sociobiology*, *62*, 1453–1466. https://doi.org/10.1007/s00265-008-0575-7

Cai, Y., Yu, H., Liu, H., Jiang, C., Sun, L., Niu, L., Liu, X., Li, D., & Li, J. (2020). Genome-wide screening of microsatellites in golden snub-nosed monkey (Rhinopithecus roxellana), for the development of a standardized genetic marker system. *Scientific Reports*, *10*. https://doi.org/10.1038/s41598-020-67451-2

Chaves, P. B., Strier, K. B., & Di Fiore, A. (2023). Paternity data reveal high MHC diversity among sires in a polygynandrous, egalitarian primate. *Proceedings of the Royal Society B*, *290*, 20231035. https://doi.org/10.1098/rspb.2023.1035

Constable, J. L., Ashley, M. V., Goodall, J., & Pusey, A. E. (2001). Noninvasive paternity assignment in Gombe chimpanzees. *Molecular Ecology*, *10*, 1279–1300. https://doi.org/10.1046/j.1365-294X.2001.01262.x

Dal Pesco, F. (2019). *Dynamics and fitness benefits of male-male sociality in wild Guinea baboons (Papio papio)* [PhD thesis]. Georg-August-Universität Göttingen.

Dal Pesco, F., Neumann, C., Trede, F., Zinner, D., & Fischer, J. (2025). Nested male reproductive strategies in a tolerant multilevel primate society. *bioRxiv*. https://doi.org/10.1101/2025.10.27.684814

Díaz-Muñoz, S. L. (2011). Paternity and relatedness in a polyandrous nonhuman primate: Testing adaptive hypotheses of male reproductive cooperation. *Animal Behaviour*, *82*, 563–571. https://doi.org/10.1016/j.anbehav.2011.06.013

Dolotovskaya, S., Roos, C., & Heymann, E. W. (2020). Genetic monogamy and mate choice in a pair-living primate. *Scientific Reports*, *10*. https://doi.org/10.1038/s41598-020-77132-9

Driller, C., Perwitasari-Farajallah, D., Zischler, H., & Merker, S. (2009). The social system of Lariang tarsiers (Tarsius lariang) as revealed by genetic analyses. *International Journal of Primatology*, *30*, 267–281. https://doi.org/10.1007/s10764-009-9341-6

Engelhardt, A., Heistermann, M., Hodges, J. K., Nürnberg, P., & Niemitz, C. (2006). Determinants of male reproductive success in wild long-tailed macaques (Macaca fascicularis)–male monopolisation, female mate choice or post-copulatory mechanisms? *Behavioral Ecology and Sociobiology*, *59*, 740–752. https://doi.org/10.1007/s00265-005-0104-x

Engelhardt, A., Muniz, L., Perwitasari-Farajallah, D., & Widdig, A. (2017). Highly polymorphic microsatellite markers for the assessment of male reproductive skew and genetic variation in critically endangered crested macaques (Macaca nigra). *International Journal of Primatology*, *38*, 672–691. https://doi.org/10.1007/s10764-017-9973-x

Feldblum, J. T., Wroblewski, E. E., Rudicell, R. S., Hahn, B. H., Paiva, T., Cetinkaya-Rundel, M., Pusey, A. E., & Gilby, I. C. (2014). Sexually coercive male chimpanzees sire more offspring. *Current Biology*, *24*, 2855–2860. https://doi.org/10.1016/j.cub.2014.10.039

Fietz, J., Zischler, H., Schwiegk, C., Tomiuk, J., Dausmann, K. H., & Ganzhorn, J. U. (2000). High rates of extra-pair young in the pair-living fat-tailed dwarf lemur, Cheirogaleus medius. *Behavioral Ecology and Sociobiology*, *49*, 8–17. https://doi.org/10.1007/s002650000269

Fox, S. et al. (n.d.). *Unpublished paternity data for ursine colobus.*

Fox, S. (2015). *The effect of potential and actual paternity on positive male-infant behaviour in ursine colobus* [Master's thesis, University of Calgary]. https://doi.org/10.11575/PRISM/28090

Gatti, S. et al. (n.d.). *Unpublished paternity data for western lowland gorillas extracted from s. Gatti 2005 PhD thesis.*

Gerloff, U., Hartung, B., Fruth, B., Hohmann, G., & Tautz, D. (1999). Intracommunity relationships, dispersal pattern and paternity success in a wild living community of Bonobos (Pan paniscus) determined from DNA analysis of faecal samples. *Proceedings of the Royal Society B*, *266*. https://doi.org/10.1098/rspb.1999.0762

Gilby, I. C., Brent, L. J. N., Wroblewski, E. E., Rudicell, R. S., Hahn, B. H., Goodall, J., & Pusey, A. E. (2013). Fitness benefits of coalitionary aggression in male chimpanzees. *Behavioral Ecology and Sociobiology*, *67*, 373–381. https://doi.org/10.1007/s00265-012-1457-6

Godoy, I., Vigilant, L., & Perry, S. E. (2016a). Cues to kinship and close relatedness during infancy in white-faced capuchin monkeys, Cebus capucinus. *Animal Behaviour*, *116*, 139–151. https://doi.org/10.1016/j.anbehav.2016.03.031

Godoy, I., Vigilant, L., & Perry, S. E. (2016b). Inbreeding risk, avoidance and costs in a group-living primate, Cebus capucinus. *Behavioral Ecology and Sociobiology*, *70*, 1601–1611. https://doi.org/10.1007/s00265-016-2168-1

Goossens, B., Setchell, J. M., James, S. S., Funk, S. M., Chikhi, L., Abulani, A., Ancrenaz, M., Lackman-Ancrenaz, I., & Bruford, M. W. (2006). Philopatry and reproductive success in Bornean orang-utans (Pongo pygmaeus). *Molecular Ecology*, *15*, 2577–2588. https://doi.org/10.1111/j.1365-294X.2006.02952.x

Guo, S., Ji, W., Li, M., Chang, H., & Li, B. (2010). The mating system of the Sichuan snub-nosed monkey (Rhinopithecus roxellana). *American Journal of Primatology*, *72*, 25–32. https://doi.org/10.1002/ajp.20747

Hayakawa, S. (2008). Male–female mating tactics and paternity of wild Japanese macaques (Macaca fuscata yakui). *American Journal of Primatology*, *70*, 986–989. https://doi.org/10.1002/ajp.20580

Heistermann, M., Ziegler, T., Schaik, C. P. van, Launhardt, K., Winkler, P., & Hodges, J. K. (2001). Loss of oestrus, concealed ovulation and paternity confusion in free-ranging Hanuman langurs. *Proceedings of the Royal Society B*. https://doi.org/10.1098/rspb.2001.1833

Higham, J. P., Heistermann, M., Agil, M., Perwitasari-Farajallah, D., Widdig, A., & Engelhardt, A. (2021). Female fertile phase synchrony, and male mating and reproductive skew, in the crested macaque. *Scientific Reports*. https://doi.org/10.1038/s41598-021-81163-1

Huang, X., Hu, N., Chapman, C. A., He, K., Jiang, X., Guan, Z., Fan, P., & Garber, P. A. (2022). Disassociation of social and sexual partner relationships in a gibbon population with stable one-male two-female groups. *American Journal of Primatology*. https://doi.org/10.1002/ajp.23394

Huchard, E., Alvergne, A., Féjan, D., Knapp, L. A., Cowlishaw, G., & Raymond, M. (2010). More than friends? Behavioural and genetic aspects of heterosexual associations in wild chacma baboons. *Behavioral Ecology and Sociobiology*, *64*, 769–781. https://doi.org/10.1007/s00265-009-0894-3

Huck, M., Fernandez-Duque, E., Babb, P., & Schurr, T. (2014). Correlates of genetic monogamy in socially monogamous mammals: Insights from Azara's owl monkeys. *Proceedings of the Royal Society B*. https://doi.org/10.1098/rspb.2014.0195

Huck, M., Löttker, P., Böhle, U.-R., & Heymann, E. W. (2005). Paternity and kinship patterns in polyandrous moustached tamarins (Saguinus mystax). *American Journal of Physical Anthropology*, *127*, 449–464. https://doi.org/10.1002/ajpa.20136

Inoue, E., Akomo-Okoue, E. F., Ando, C., Iwata, Y., Judai, M., Fujita, S., Hongo, S., Nze-Nkogue, C., Inoue-Murayama, M., & Yamagiwa, J. (2013). Male genetic structure and paternity in western lowland gorillas (gorilla gorilla gorilla). *American Journal of Physical Anthropology*, *151*, 583–588. https://doi.org/10.1002/ajpa.22312

Inoue, E., Inoue-Murayama, M., Vigilant, L., Takenaka, O., & Nishida, T. (2008). Relatedness in wild chimpanzees: Influence of paternity, male philopatry, and demographic factors. *American Journal of Physical Anthropology*, *137*, 256–262. https://doi.org/10.1002/ajpa.20865

Inoue, E., & Takenaka, O. (2008). The effect of male tenure and female mate choice on paternity in free-ranging japanese macaques. *American Journal of Primatology*, *70*, 62–68. https://doi.org/10.1002/ajp.20457

Ishizuka, S., Kawamoto, Y., Sakamaki, T., Tokuyama, N., Toda, K., Okamura, H., & Furuichi, T. (2018). Paternity and kin structure among neighbouring groups in wild bonobos at wamba. *Royal Society Open Science*, *5*, 171006. https://doi.org/10.1098/rsos.171006

Jack, K. M., & Fedigan, L. M. (2006). Why be alpha male? Dominance and reproductive success in wild white-faced capuchins (cebus capucinus). In *New perspectives in the study of mesoamerican primates* (pp. 367–386). Springer. https://doi.org/10.1007/0-387-25872-8_18

Jacobs, R. L., Frankel, D. C., Rice, R. J., Kiefer, V. J., & Bradley, B. J. (2018). Parentage complexity in socially monogamous lemurs (eulemur rubriventer): Integrating genetic and observational data. *American Journal of Primatology*. https://doi.org/10.1002/ajp.22738

Kappeler, P. M., & Port, M. (2008). Mutual tolerance or reproductive competition? Patterns of reproductive skew among male redfronted lemurs (eulemur fulvus rufus). *Behavioral Ecology and Sociobiology*, *62*, 1477–1488. https://doi.org/10.1007/s00265-008-0577-5

Kappeler, P. M., & Schäffler, L. (2008). The lemur syndrome unresolved: Extreme male reproductive skew in sifakas (propithecus verreauxi), a sexually monomorphic primate with female dominance. *Behavioral Ecology and Sociobiology*, *62*, 1007–1015. https://doi.org/10.1007/s00265-007-0528-6

Keane, B., Dittus, W. P. J., & Melnick, D. J. (1997). Paternity assessment in wild groups of toque macaques macaca sinica at polonnaruwa, sri lanka using molecular markers. *Molecular Ecology*, *6*, 267–282. https://doi.org/10.1046/j.1365-294x.1997.00178.x

Kenyon, M., Roos, C., Binh, V. T., & Chivers, D. (2011). Extrapair paternity in golden-cheeked gibbons (nomascus gabriellae) in the secondary lowland forest of cat tien national park, vietnam. *Folia Primatologica*, *82*, 154–164. https://doi.org/10.1159/000333143

Kümmerli, R., & Martin, R. D. (2005). Male and female reproductive success in macaca sylvanus in gibraltar: No evidence for rank dependence. *International Journal of Primatology*, *26*(6), 1229–1247. https://doi.org/10.1007/s10764-005-8851-0

Langergraber, K. E., Mitani, J. C., Watts, D. P., & Vigilant, L. (2013). Male–female socio-spatial relationships and reproduction in wild chimpanzees. *Behavioral Ecology and Sociobiology*, *67*, 861–873. https://doi.org/10.1007/s00265-013-1509-6

Larney, E. (2013). *The influence of genetic and social structure on reproduction in phayre's leaf monkeys (trachypithecus phayrei crepusculus)* [PhD thesis]. Stony Brook University.

Launhardt, K., Borries, C., Hardt, C., Epplen, J. T., & Winkler, P. (2001). Paternity analysis of alternative male reproductive routes among the langurs (semnopithecus entellus) of ramnagar. *Animal Behaviour*, *61*, 53–64. https://doi.org/10.1006/anbe.2000.1590

Lawler, R. R. (2007). Fitness and extra-group reproduction in male verreaux's sifaka: An analysis of reproductive success from 1989–1999. *American Journal of Physical Anthropology*, *132*, 267–277. https://doi.org/10.1002/ajpa.20507

Lawler, R. R., Richard, A. F., & Riley, M. A. (2003). Genetic population structure of the white sifaka (Propithecus verreauxi verreauxi) at Beza Mahafaly Special Reserve, southwest Madagascar (1992–2001). *Molecular Ecology*, *12*, 2307–2317. https://doi.org/10.1046/j.1365-294X.2003.01909.x

Liu, Z., Huang, C., Zhou, Q., Li, Y., Wang, Y., Li, M., Takenaka, O., & Takenaka, A. (2013). Fecal sample collection and genetic analysis of white-headed langurs. *Integrative Zoology*, *8*, 410–416. https://doi.org/10.1111/1749-4877.12048

Masi, S., Austerlitz, F., Chabaud, C., Lafosse, S., Marchi, N., Georges, M., Dessarps-Freichey, F., Miglietta, S., Sotto-Mayor, A., San Galli, A., Meulman, E., Pouydebat, E., Krief, S., Todd, A., Fuh, T., Breuer, T., & Ségurel, L. (2021). No evidence for female kin association, indications for extragroup paternity, and sex-biased dispersal patterns in wild western gorillas. *Ecology and Evolution*. https://doi.org/10.1002/ece3.7596

McCarthy, M. S., Lester, J. D., Cibot, M., & McLennan, M. R. (2020). Atypically high reproductive skew in a small wild chimpanzee community in a human-dominated landscape. *Folia Primatologica*, *91*, 688–696. https://doi.org/10.1159/000508609

Miller, C. M., Snyder-Mackler, N., Nguyen, N., Fashing, P. J., Tung, J., Wroblewski, E. E., Gustison, M. L., & Wilson, M. L. (2021). Extragroup paternity in gelada monkeys, Theropithecus gelada, at Guassa, Ethiopia and a comparison with other primates. *Animal Behaviour*, *177*, 277–301. https://doi.org/10.1016/j.anbehav.2021.05.008

Minker, M. M. I., Young, C., Amici, F., McFarland, R., Barrett, L., Grobler, J. P., Henzi, S. P., & Widdig, A. (2018). Assessment of male reproductive skew via highly polymorphic STR markers in wild vervet monkeys, Chlorocebus pygerythrus. *Journal of Heredity*, *109*, 780–790. https://doi.org/10.1093/jhered/esy048

Moscovice, L. R., Di Fiore, A., Crockford, C., Kitchen, D. M., Wittig, R., Seyfarth, R. M., & Cheney, D. L. (2010). Hedging their bets? Male and female chacma baboons form friendships based on likelihood of paternity. *Animal Behaviour*, *79*, 1007–1015. https://doi.org/10.1016/j.anbehav.2010.01.013

Moscovice, L., Seyfarth, R., & Silk, J. (n.d.). *Unpublished paternity data for chacma baboons*.

Mouginot, M., Cheng, L., Wilson, M. L., Feldblum, J. T., Städele, V., Wroblewski, E. E., Vigilant, L., Hahn, B. H., Li, Y., Gilby, I. C., Pusey, A. E., & Surbeck, M. (2023). Reproductive inequality among males in the genus Pan. *Philosophical Transactions of the Royal Society B*, *378*, 20220301. https://doi.org/10.1098/rstb.2022.0301

Muniz, L., Perry, S., Manson, J. H., Gilkenson, H., Gros-Louis, J., & Vigilant, L. (2010). Male dominance and reproductive success in wild white-faced capuchins (Cebus capucinus) at Lomas Barbudal, Costa Rica. *American Journal of Primatology*, *72*, 1118–1130. https://doi.org/10.1002/ajp.20876

Newton-Fisher, N. E., Emery Thompson, M., Reynolds, V., Boesch, C., & Vigilant, L. (2010). Paternity and social rank in wild chimpanzees (Pan troglodytes) from the Budongo Forest, Uganda. *American Journal of Physical Anthropology*, *142*, 417–428. https://doi.org/10.1002/ajpa.21241

Nievergelt, C. M., Mutschler, T., Feistner, A. T. C., & Woodruff, D. S. (2002). Social system of the Alaotran Gentle Lemur (Hapalemur griseus alaotrensis): Genetic Characterization of Group Composition and Mating System. *American Journal of Primatology*, *57*, 157–176. https://doi.org/10.1002/ajp.10046

Nsubaga, A. M., Robbins, M. M., Boesch, C., & Vigilant, L. (2008). Patterns of paternity and group fission in wild multimale mountain gorilla groups. *American Journal of Physical Anthropology*, *135*, 263–274. https://doi.org/10.1002/ajpa.20740

Ohsawa, H., Inoue, M., & Takenaka, O. (1993). Mating strategy and reproductive success of male Patas Monkeys (Erythrocebus patas). *Primates*, *34*(4), 533–544. https://doi.org/10.1007/BF02382664

Oka, T., & Takenaka, O. (2001). Wild Gibbons' Parentage Tested by Non-invasive DNA Sampling and PCR-amplified Polymorphic Microsatellites. *Primates*, *42*(1), 67–73. https://doi.org/10.1007/BF02640690

Oklander, L. I., Kowalewski, M., & Corach, D. (2014). Male reproductive strategies in black and gold howler monkeys (Alouatta caraya). *American Journal of Primatology*, *76*, 43–55. https://doi.org/10.1002/ajp.

22191

Parga, J. A., Sauther, M. L., Cuozzo, F. P., Jacky, I. A. Y., Lawler, R. R., Sussman, R. W., Gould, L., & Pastorini, J. (2016). Paternity in wild ring-tailed lemurs (Lemur catta): Implications for male mating strategies. *American Journal of Primatology*, *78*(12), 1316–1325. https://doi.org/10.1002/ajp.22584

Platner, B. (2005). *Das genetische Paarungssystem von Lepilemur ruficaudatus* [Master's thesis]. University of Zurich.

Pope, T. R. (1990). The reproductive consequences of male cooperation in the red howler monkey: Paternity exclusion in multi-male and single-male troops using genetic markers. *Behavioral Ecology and Sociobiology*, *27*(6), 439–446. https://doi.org/10.1007/BF00164071

Qi, X.-G., Grueter, C. C., Fang, G., Huang, P.-Z., Zhang, J., Duan, Y.-M., Huang, Z.-P., Garber, P. A., & Li, B.-G. (2020). Multilevel societies facilitate infanticide avoidance through increased extrapair matings. *Animal Behaviour*, *161*, 127–137. https://doi.org/10.1016/j.anbehav.2019.12.014

Roberts, S.-J., Nikitopoulos, E., & Cords, M. (2014). Factors affecting low resident male siring success in one-male groups of blue monkeys. *Behavioral Ecology*, *25*(4), 852–861. https://doi.org/10.1093/beheco/aru060

Rosenbaum, S., Hirwa, J. P., Silk, J. B., Vigilant, L., & Stoinski, T. S. (2015). Male rank, not paternity, predicts male–immature relationships in mountain gorillas, Gorilla beringei beringei. *Animal Behaviour*, *104*, 13–24. https://doi.org/10.1016/j.anbehav.2015.03.001

Ruiter, J. R. de, Hooff, J. A. R. A. M. van, & Scheffrahn, W. (1994). Social and genetic aspects of paternity in wild long-tailed macaques (Macaca fascicularis). *Behaviour*, *129*(3/4), 203–224. https://doi.org/10.1163/156853994X00613

Schülke, O., Kappeler, P. M., & Zischler, H. (2004). Small testes size despite high extra-pair paternity in the pair-living nocturnal primate Phaner furcifer. *Behavioral Ecology and Sociobiology*, *55*, 293–301. https://doi.org/10.1007/s00265-003-0709-x

Schwensow, N., Fietz, J., Dausmann, K., & Sommer, S. (2008). MHC-associated mating strategies and the importance of overall genetic diversity in an obligate pair-living primate. *Evolutionary Ecology*, *22*, 617–636. https://doi.org/10.1007/s10682-007-9186-4

Scott, A. M., Banes, G. L., Saragih, J. R., & Knott, C. D. (2024). Flanged males have higher reproductive success in a completely wild orangutan population. *PLOS ONE*, *19*(2), e0296688. https://doi.org/10.1371/journal.pone.0296688

Silk, J. et al. (n.d.). *Unpublished paternity data for olive baboons*.

Snyder-Mackler, N., Alberts, S. C., & Bergman, T. J. (2012). Concessions of an alpha male? Cooperative defence and shared reproduction in multi-male primate groups. *Proceedings of the Royal Society B*, *279*, 3788–3795. https://doi.org/10.1098/rspb.2012.0842

Soltis, J., Thomsen, R., & Takenaka, O. (2001). The interaction of male and female reproductive strategies and paternity in wild Japanese macaques, Macaca fuscata. *Animal Behaviour*, *62*, 485–494. https://doi.org/10.1006/anbe.2001.1774

Strier, K. B., Chaves, P. B., Mendes, S. L., Fagundes, V., & Di Fiore, A. (2011). Low paternity skew and the influence of maternal kin in an egalitarian, patrilocal primate. *Proceedings of the National Academy of Sciences*, *108*(45), 17884–17889. https://doi.org/10.1073/pnas.1116737108

Sukmak, M., Wajjwalku, W., Ostner, J., & Schülke, O. (2014). Dominance rank, female reproductive synchrony, and male reproductive skew in wild Assamese macaques. *Behavioral Ecology and Sociobiology*, *68*, 1097–1108. https://doi.org/10.1007/s00265-014-1721-z

Surbeck, M., Langergraber, K. E., Fruth, B., Vigilant, L., & Hohmann, G. (2017). Male reproductive skew is higher in bonobos than chimpanzees. *Current Biology*, *27*(13), R503–R505. https://doi.org/10.1016/j.cub.2017.05.039

Tajima, T., Malim, T. P., & Inoue, E. (2018). Reproductive success of two male morphs in a free-ranging population of Bornean orangutans. *Primates*, *59*, 127–133. https://doi.org/10.1007/s10329-017-0648-1

Teichroeb, J. A., Wikberg, E. C., Ting, N., & Sicotte, P. (2014). Factors influencing male affiliation and coalitions in a species with male dispersal and intense male–male competition, Colobus vellerosus. *Behaviour*, *151*(7), 1045–1066. https://doi.org/10.1163/1568539X-00003089

Trébouet, F. (2019). *Male reproductive strategies in wild northern Pig-Tailed Macaques (Macaca leonina): Testing the priority-of-access model* [PhD thesis]. Southern Illinois University Carbondale.

Utami, S. S., Goossens, B., Bruford, M. W., Ruiter, J. R. de, & Hooff, J. A. R. A. M. van. (2002). Male bimaturism and reproductive success in Sumatran orang-utans. *Behavioral Ecology*, *13*(5), 643–652. https://doi.org/10.1093/beheco/13.5.643

Van Belle, S., Estrada, A., & Strier, K. B. (2016). Social relationships and the use of space in Alouatta pigra. *International Journal of Primatology*, *37*, 441–455. https://doi.org/10.1007/s10764-016-9905-1

Vigilant, L., Hofreiter, M., Siedel, H., & Boesch, C. (2001). Paternity and relatedness in wild chimpanzee communities. *Proceedings of the National Academy of Sciences*, *98*(23), 12890–12895. https://doi.org/10.1073/pnas.231320498

Vigilant, L., Roy, J., Bradley, B. J., Stoneking, C. J., Robbins, M. M., & Stoinski, T. S. (2015). Reproductive competition and inbreeding avoidance in a primate species with habitual female dispersal. *Behavioral Ecology and Sociobiology*, *69*, 1163–1172. https://doi.org/10.1007/s00265-015-1930-0

Walker, K. K., Rudicell, R. S., Li, Y., Hahn, B. H., Wroblewski, E., & Pusey, A. E. (2017). Chimpanzees breed with genetically dissimilar mates. *Royal Society Open Science*, *4*, 160422. https://doi.org/10.1098/rsos.160422

Wang, B.-S., Wang, Z.-L., Tian, J.-D., Cui, Z.-W., & Lu, J.-Q. (2015). Establishment of a microsatellite set for noninvasive paternity testing in free-ranging Macaca mulatta tcheliensis in Mount Taihangshan area, Jiyuan, China. *Zoological Studies*, *54*, 8. https://doi.org/10.1186/s40555-014-0100-9

Weiss, A., Feldblum, J. T., Altschul, D. M., Collins, D. A., Kamenya, S., Mjungu, D., Foerster, S., Gilby, I. C., Wilson, M. L., & Pusey, A. E. (2023). Personality traits, rank attainment, and siring success throughout the lives of male chimpanzees of Gombe National Park. *PeerJ*, *11*, e15083. https://doi.org/10.7717/peerj.15083

Wikberg, E. C., Jack, K. M., Fedigan, L. M., Campos, F. A., Yashima, A. S., Bergstrom, M. L., Hiwatashi, T., & Kawamura, S. (2017). Inbreeding avoidance and female mate choice shape reproductive skew in capuchin monkeys (Cebus capucinus imitator). *Molecular Ecology*, *26*, 653–667. https://doi.org/10.1111/mec.13898

Wimmer, B., & Kappeler, P. M. (2002). The effects of sexual selection and life history on the genetic structure of redfronted lemur, Eulemur fulvus rufus, groups. *Animal Behaviour*, *64*, 557–568. https://doi.org/10.1006/anbe.2002.4003

Wroblewski, E. E., Murray, C. M., Keele, B. F., Schumacher-Stankey, J. C., Hahn, B. H., & Pusey, A. E. (2009). Male dominance rank and reproductive success in chimpanzees, Pan troglodytes schweinfurthii. *Animal Behaviour*, *77*(4), 873–885. https://doi.org/10.1016/j.anbehav.2008.12.014

Wu, F., Liu, J., Dunn, D. W., Shang, Y., Jin, S., Du, H., Wu, Y., Men, Y., Chen, G., He, G., Li, B., & Guo, S. (2025). Beyond the alpha: Extra-pair paternities and male reproductive success in a primate multilevel society. *Ecology and Evolution*, *15*, e70928. https://doi.org/10.1002/ece3.70928

Xiang, Z.-F., Yang, B.-H., Yu, Y., Yao, H., Grueter, C. C., Garber, P. A., & Li, M. (2014). Males collectively defend their one-male units against bachelor males in a multi-level primate society. *American Journal of Primatology*, *76*, 609–617. https://doi.org/10.1002/ajp.22254

Yamane, A., Shotake, T., Mori, A., Boug, A. I., & Iwamoto, T. (2003). Extra-unit paternity of hamadryas baboons (Papio hamadryas) in Saudi Arabia. *Ethology Ecology and Evolution*, *15*(4), 379–387. https://doi.org/10.1080/08927014.2003.9522664

Yang, B.-H., Ren, B., Xiang, Z.-F., Yang, J., Yao, H., Garber, P. A., & Li, M. (2014). MHC-DRB exon II variation in a wild population of Rhinopithecus roxellana. *Integrative Zoology*, *9*, 598–612. https://doi.org/10.1111/1749-4877.12084